Fast Fingerprint Authentication Based on Ultrasonic Guided Waves

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***Abstract*—Fingerprint scanning is a biometric technology widely used in various fields, such as mobile payment and intelligent security, due to its high accuracy and reliability. Compared with capacitive and optical fingerprint scanning, ultrasonic fingerprint scanning can fix the authentication issue when fingertips are wet, oily or dusty. However, achieving fast and accurate ultrasonic fingerprint scanning and authentication in a large physical domain is challenging. A novel technique, combining deep learning and tomography, is suggested by this study to tackle these matters. In this study, fingerprint reconstruction at any position is achieved in a large physical domain by monitoring the wave field changes of a plate-like structure using ultrasonic guided waves. Accurately characterization and authentication of the fingerprint can be achieved through the utilization of fast inversion tomography (FIT) and mask region convolutional neural network (Mask R-CNN). A sequence of data and image processing shows that FIT can effectively characterize fingerprint features in the array, and the post-processing method based on the Mask R-CNN model can achieve fast and accurate ultrasonic fingerprint matching in a large physical domain. This further enhances the comfort and security of devices that use fingerprint identification.**

Keywords—Fingerprint Authentication, Ultrasonic Guided Waves, Fast Inversion Tomography, Mask Region Convolutional Neural Network

I. Introduction

Fingerprints, voiceprints, palmprints, irises, and other biometric features are unique characteristics found on every individual. Due to their difficulty to forge or alter, biometric features have been widely applied in fields such as identity verification, security measures, and crime detection [1]. Biometric recognition is the process of acquiring, extracting, and comparing an individual's biometric features using computer technology [2]. Among them, fingerprint authentication has become the most common and reliable biometric recognition technology in daily life due to its lifelong stability, ease of collection, and high recognition accuracy. It is widely used in various fields such as unlocking smartphones and mobile payments.

Since the 20th century, modern fingerprint detection technology has evolved from optical to capacitive-based methods. In recent years, ultrasound-based fingerprint detection technology has been gaining traction due to advances in technology. Optical and capacitive fingerprint detection techniques are straightforward to acquire and authenticate fingerprint images, but can be affected by contaminants like water, oil, dirt, and other substances on the finger's surface, which can compromise authentication accuracy [3]. This technology is unaffected by the humidity, oil, or other pollutants on the fingerprint surface, enabling accurate capture of fingerprint information [4]. However, the accuracy of ultrasound-based fingerprint reconstruction is somewhat compromised due to limitations in sensor resolution, signal attenuation during propagation, and interference from environmental noise.

The traditional technique of ultrasonic fingerprint detection relies on pulse echo to capture fingerprint information, which imposes certain limitations on the size of the fingerprint scanning physical domain. In previous studies, the variation of the wavefield on a glass plate was detected using ultrasonic guided waves, enabling the reconstruction of fingerprints at arbitrary positions within a sensor array. By combining deep learning convolutional neural networks (DLCNN) with fast inversion tomography (FIT) and optimizing the parameters, a quantitative representation of fingerprints can be obtained, revealing the details of sub-millimeter fingerprints and eliminating reconstruction artifacts, thereby significantly improving the imaging quality and resolution [5].

Traditional fingerprint authentication techniques are mainly based on the analysis and matching of minutia features such as endpoints and bifurcation points in fingerprint ridges [6]. However, traditional methods often only take advantage of local fingerprints and their features when extracting and storing minutiae, such as singular points, which can lead to an inadequate and error detection of these features and errors in detection. Additionally, increasing the size of the fingerprint database and comparing multiple feature fingerprints consumes a significant amount of time, making real-time authentication difficult to achieve. To address the aforementioned issues, it is necessary to consider improving the method of extracting minutiae, exploring new fingerprint feature information for fingerprint authentication, and

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adopting faster and more accurate fingerprint acquisition and authentication methods.

The accuracy and robustness of algorithms in various fields have been significantly enhanced by the widespread application of deep learning and artificial intelligence technologies in recent years. Mask region convolutional neural network (MASK R-CNN), as a hotspot in instance segmentation technology, is of particular interest in this work, due to its integration with FIT for fingerprint acquisition and authentication [7]. Thus, in order to emphasize the intelligence of rapid fingerprint authentication, the DLCNN method used for artifact removal is temporarily disregarded, and instead, a binarization filtering method is directly employed for fingerprint image normalization. Ultimately, the reconstructed fingerprint in binary form is used for determination, segmentation, and authentication of minutiae.

II. Method

A. Forward Problem

Based on the nature of the plate-like structure, the distribution of the wave field can be obtained through forward modeling using the wave equation. By employing the Fourier transform to the two-dimensional acoustic wave equation of isotropic media in the time-space domain, the following can be derived:

$$-\omega^2 Mu(x,\omega) + i\omega pu(x,\omega) + Ku(x,\omega) = s(x,\omega) \quad (1)$$

where ω denotes the angular frequency, x denotes the spatial coordinates, $u(x,\omega)$ denotes the pressure wave field, M refers to the mass matrix, K denotes the stiffness matrix, and $s(x,\omega)$ denotes the point source term. Furthermore, $pu(x,\omega)$ denotes the additional damping term introduced when considering the absorption boundary condition, and p is the damping coefficient [8].

The finite element finite difference technique is the most widely employed approach to resolving the two-dimensional acoustic wave equation in isotropic media, both in terms of time and frequency [9]. In this method, all the model parameters, including in (1), are taken into account.

B. Supervised Descent Method and Optimal Model

Two components make up the FIT technique: ascertaining the descent direction matrix and forecasting the ideal wavefield model. The wavefield prediction involves iterative processes to determine an optimal model that best explains the observed wavefield. During the iterative process, the traditional full waveform inversion method for solving for the Jacobian matrix J and the Hessian matrix H requires significant computational resources. Hence, the supervised descent method is employed to calculate the descent direction matrix R instead of explicitly computing these matrices [10].

After n iterations, the minimum cost function can be expressed as follows:

$$\arg\min_{R_n} \left\| \Delta M_n - \Delta V_n \; R_n \; \right\|_F^2 + q \left\| R_n \right\|_F^2 \quad (2)$$

where $\Delta M_n = M_{true} - M_n$, $\Delta V_n = V_{true} - V_n$, M_{true} denotes the slowness matrix, V_{true} denotes the frequency domain matrix, M_n and V_n correspond to the matrices after n iterations. q denotes the regularization weight, ensuring the stability of the iterative process, while F denotes the Frobenius norm. By solving (2), the descent gradient R_n can be obtained.

Through inversion, unknown models can be directly predicted. If the result of the nth iteration R_n denotes a generally descending direction that reaches the threshold, then after the nth iteration, the model is k_{n+1}:

$$k_{n+1} = k_n + \left(d_n^{pred} - d^{obs} \right) R_n = k_n + \Delta d_n R_n \quad (3)$$

where, d_n^{pred} denotes the predicted wavefield and d^{obs} corresponds to the observed wavefield at the nth iteration.

C. Image Filtering

Gabor filter exhibits excellent frequency and orientation selectivity, providing the best joint resolution in both space and frequency domains [11]. As a result, Gabor filters are often employed to eliminate noise and guarantee the precise portrayal of valleys and ridges in fingerprint minutiae [12].

In the spatial domain, an even-symmetric Gabor filter is typically expressed as such:

$$G(x,y,\theta,f) = \exp\left\{ -\frac{1}{2}\left[\frac{x_\theta^{\,2}}{\sigma_x^{\,2}} + \frac{y_\theta^{\,2}}{\sigma_y^{\,2}} \right] \right\} \cos\left(2\pi f x_\theta\right) \quad (4)$$

where $\begin{cases} x_\theta = x cos\theta + y sin\theta \\ y_\theta = -x sin\theta + y cos\theta \end{cases}$, θ denotes the orientation of the Gabor filter, f is the frequency of the sinusoidal plane wave, σ_x and σ_y are the Gaussian envelope constants along the x and y axes, respectively. In fingerprint image filtering, θ and f are associated with the local ridge direction and frequency, determining their spatial position.

Convoluting the Gabor filter with the image yields the filtered image. By setting a threshold value, each pixel can be converted into a binary image with pure black or white.

III. Numerical Research and Results

A. Material and Guided Wave Dispersion

TABLE I reveals the material parameters of the finger epidermis and glass plate during the investigation of linear sound fields. ρ is the mass density, λ and μ are the Lame constants [13,14]. When a finger is placed on the glass plate, it causes significant changes in the wave field of the glass plate, as illustrated in Fig. 1. Fig. 2 displays the phase velocity dispersion curve of the Corning Gorilla Glass plate, which was determined through the global matrix method. The black curve in the figure symbolizes the glass plate's phase velocity dispersion when the finger is pressed. To guarantee reconstruction accuracy and minimize multi-mode interference, the A_0 mode at $fd = 2.25 MHz \cdot mm$ of excitation is chosen, and the corresponding phase velocity value c_p of the A_0 mode is used as the basis for the

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velocity map. It forms the range of c_p variation after the fingerprint pressing, in conjunction with adjacent curves.

TABLE I. MATERIAL PARAMETERS OF FINGERS AND GLASS

Material	Parameters and Units		
	ρ / gcm^{-3}	λ / GPa	μ / GPa
Cuticles of Fingers	1.02	3.13	0.78
Corning Gorilla Glass 6	2.4	23.04	31.82

Fig. 1. Schematic diagram of sound field changes.

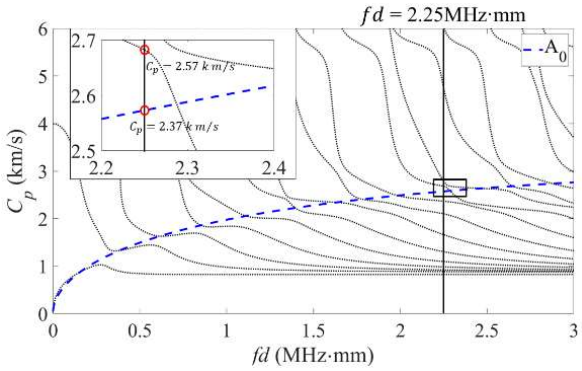

Fig. 2. Phase velocity dispersion curve.

B. Finite Difference Model

In the research of the finite-difference model, the finite-difference parameters are shown in TABLE II. A circular array of 128 elements with a diameter of 40mm is placed at the center of a 50×50mm glass plate. A grid of 500×500 points, with a spacing of 0.1mm, forms the finite-difference model. By using c_p curves, the glass plate and fingerprint model are reduced to a velocity model and subjected to sound field simulation. Fig. 5(a) displays the outcomes.

TABLE II. FINITE DIFFERENCE PARAMETERS

Number of Array Elements	*Array Diameter*	*Glass Panel Size*	*Grid Spacing*	*Number of Grids*
128	40mm	50×50×0.5mm	0.1mm	500×500

C. MASK R-CNN Model

An algorithm that fuses object detection and semantic segmentation, Mask R-CNN is composed of four main components: Backbone, Region Proposal Network (RPN), Region of Interest (ROI) Align, and Fully Convolution Network (FCN). The network architecture is illustrated in Fig. 3. To begin, the Backbone functions as a feature extraction network to draw out and combine the feature data from the input images. Subsequently, the RPN network initially chooses the areas where the objects are probable to be present. Next, ROI Align is utilized to extract features from each region of interest, resulting in a more refined feature map. At long last, FCN classifies and regresses the potential areas to acquire the bounding boxes with the objects of interest, as well as carrying out the segmentation and masking of the items within the bounding boxes.

The annotated fingerprint images are inputted into Mask R-CNN for training iterations, including 50, 100, 150, and 200 iterations. In particular, the loss for iterations 50 and 100 is higher compared to that of iteration 150, while the loss for iterations 150 and 200 is both about 0.13. However, iteration 200 takes approximately 6 more hours than iteration 150. Therefore, it is decided to perform 150 iterations. Fig. 4 displays the loss curve through the iterations, and the results of fingerprint segmentation and authentication are shown in Fig. 6(d).

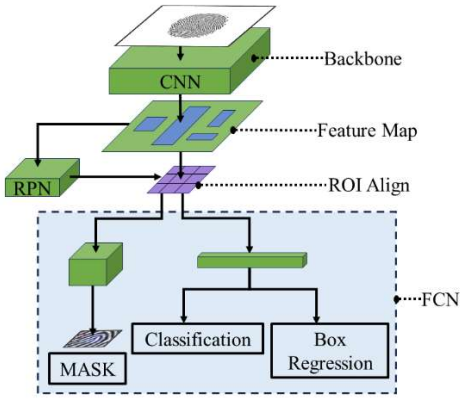

Fig. 3. MASK R-CNN network diagram.

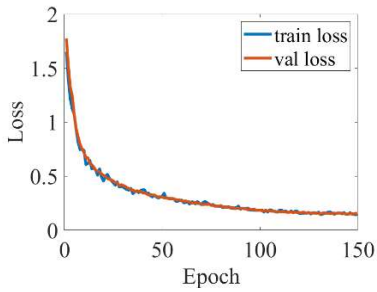

Fig. 4. Loss descent curve.

D. Results

The fingerprint true model obtained through finite-difference modeling is shown in Fig. 5(a). The fingerprint reconstructed by the FIT algorithm based on supervised descent method is shown in Fig. 5(b). The Pearson coefficient between Fig. 5(a) and 5(b) is 0.75. Converting the reconstructed result into a grayscale image yields Fig. 5(c). Ultrasonic guided wave fingerprint imaging has the advantage of large physical domain reconstruction. However, from Fig. 5(b) and 5(c), it can be observed that there are certain artifacts in the reconstructed result, indicating that the inversion accuracy is not as good as the traditional fingerprint imaging technique based on pulse echo, which significantly affects the quality of fingerprint imaging and the accuracy of fingerprint matching. Therefore, filtering the grayscale image with Gabor filters and performing image binarization to eliminate artifacts results in Fig. 5(d), thereby improving the image quality of the reconstructed result.

Traditional fingerprint annotation based on minutiae is shown in Fig. 6(a), and the fingerprint authentication results along with their confidences are shown in Fig. 6(b). The traditional fingerprint authentication method takes 16.08

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seconds per single core run, with a fingerprint matching rate of 62.69%. The traditional fingerprint authentication methods primarily involve annotating and matching the termination points and bifurcation points of fingerprint ridges. However, when there are defects in image quality or other factors affecting the characterization of termination points and bifurcation points, it can easily impact the accuracy of final fingerprint matching and authentication. Fingerprint annotation based on MASK R-CNN is shown in Fig. 6(c), while the fingerprint authentication results and confidences are shown in Fig. 6(d). The fingerprint authentication method based on MASK R-CNN takes 2.78 seconds per single core run, with a fingerprint matching rate of 99.30%. On the other hand, the fingerprint authentication method based on MASK R-CNN relies on annotating and matching feature regions that contain multiple fine nodes, along with their relative positions and ridge orientations. Even if the characterization of a certain fine node is not ideal, it does not affect the overall fingerprint matching process, allowing for more accurate fingerprint authentication. When compared to the customary fingerprint authentication technique, MASK R-CNN-based fingerprint authentication accelerates recognition by 4.78 times and augments the matching rate by 36.61% significantly. From the above data, it can be concluded that the traditional fingerprint authentication method has strict requirements for fingerprint images and lacks robustness compared to intelligent algorithms such as MASK R-CNN.

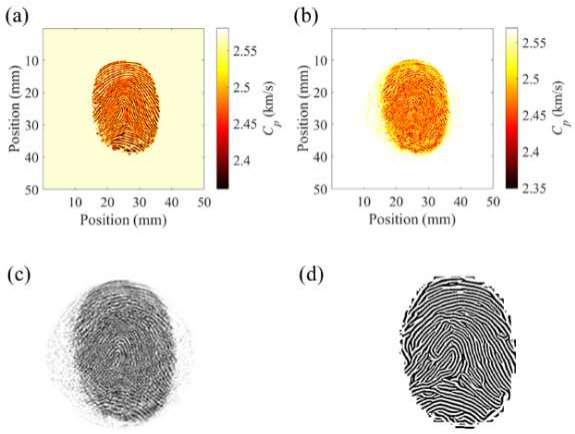

Fig. 5. (a) Real fingerprint model. (b) Fingerprint reconstruction results. (c) Grayscale image of reconstruction results. (d) Grayscale image filtering results.

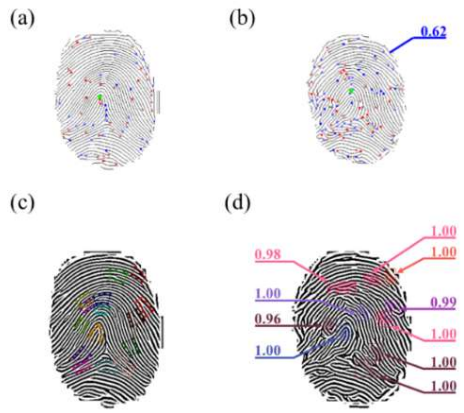

Fig. 6. (a) Traditional fingerprint annotation. (b) Traditional fingerprint authentication results. (c) MASK R-CNN fingerprint annotation. (d) MASK R-CNN authentication results.

IV. Conclusion

This work focuses on the research of fast fingerprint authentication technology based on ultrasonic guided waves. By utilizing the FIT algorithm, fingerprint images are rapidly reconstructed in a large physical domain, improving the ultrasonic fingerprint scanning technique. Standardization of fingerprint images is performed directly through image filtering and binarization operations, replacing the DLCNN artifact removal algorithm used in previous work. The algorithm's complexity is simplified and the image quality of fingerprint reconstruction is improved by this. Finally, the MASK R-CNN algorithm is employed for direct segmentation and authentication on the standardized images, enhancing the intelligence of fingerprint authentication technology. In summary, the fusion of ultrasonic guided wave-based reconstruction and the MASK R-CNN algorithm in fingerprint authentication methods not only enables fast fingerprint authentication in a large physical domain but also offers high accuracy.

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